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# Navigating the Yield Curve by the Stars

**Abstract**

Most term structure models assume constant long-term means, despite the pronounced secular decline in interest rates over recent decades. We extend an arbitrage-free Nelson-Siegel model of nominal and real yields by incorporating shifting endpoints, allowing the long-run anchors of the yield curve to evolve over time with underlying macroeconomic trends. Its performance is then evaluated relative to the fixed-endpoint counterpart using both in-sample fit and forecasting accuracy. The results show that allowing for time variation in unconditional means improves model fit and reveals negative co-movement between the slope factor and the transitory part of the level factor. Out-of-sample, the shifting-endpoint extension consistently delivers lower root mean squared forecast errors, with gains of approximately 19 basis points for nominal yields and 34 basis points for real yields at the two-year horizon. Improvements are already present at short horizons and become more pronounced as the horizon increases, highlighting the importance of accounting for shifting long-run expectations.
**Keywords:** Term structure modeling, Shadow rate, Shifting endpoints

# 1 Introduction

Over the past decades, nominal and real interest rates have experienced a structural decline, driven by changes in underlying macroeconomic conditions. However, most term structure models assume constant long-term means and do not accommodate this downward trend in interest rates. Structural changes are therefore captured as temporary fluctuations around a constant mean, rather than as shifts in the long-run component of the yield curve. Allowing for time-varying long-term means, commonly referred to as shifting endpoints, can address this issue. Distinguishing between long-run trends and cyclical variation is important for understanding the drivers of interest rate movements. It tells policymakers whether a change in interest rates reflects a shift in economic fundamentals or a temporary fluctuation, and it gives investors a better basis for interpreting yield curve dynamics and forecasting.

Several contributions have introduced shifting endpoints into term structure models for nominal yields, but important limitations remain. Within the Nelson-Siegel literature, Van Dijk et al. (2014) and Opschoor and Van der Wel (2025) implement shifting endpoints in dynamic and smooth shadow-rate specifications, retaining the interpretability of the factors but without imposing no-arbitrage restrictions. In no-arbitrage frameworks, Kozicki and Tinsley (2001) and Cieslak and Povala (2015) show that shifting long-run inflation expectations account for the persistent component of yields and improve yield forecasts. Building on these insights, Bauer and Rudebusch (2020) show that both long-run inflation expectations and the equilibrium real interest rate are necessary to capture the trend in nominal yields, thereby improving term premium estimates and out-of-sample forecasting performance. However, as their analysis focuses on nominal yields, the equilibrium real rate and long-run inflation expectations cannot be identified separately, but only through their combined effect.

We contribute to the literature by introducing a term structure model with shifting endpoints that jointly models nominal and real yields. As a result, the equilibrium real rate and long-run inflation expectations can be identified separately. In this context, our central research question is whether incorporating shifting endpoints improves the in-sample fit and leads to more accurate forecasts. We address this question by extending the five-factor model of Andreasen et al. (2021) with shifting endpoints to capture trends in long-run means. To the best of our knowledge, this is the first study to introduce shifting endpoints within the arbitrage-free Nelson-Siegel (AFNS) framework of Christensen et al. (2011), and to produce forecasts for real yields.

Incorporating shifting endpoints improves both in-sample fit and out-of-sample performance for nominal and real yields. The shifting-endpoint specification achieves more accurate pricing and is favored by all information criteria. Furthermore, the cyclical component of the level factor exhibits a strong negative relationship with the slope factor, capturing variation linked to macroeconomic events. Out-of-sample, the shifting-endpoint specification reduces forecast errors relative to the fixed-endpoint model by 19 basis points for ten-year nominal yields and 34 basis points for ten-year real yields at the two-year horizon. The gains become larger at longer horizons, reflecting the increasing importance of mean reversion toward a correctly specified long-run anchor.

The remainder of this paper is structured as follows. Section 2 describes the yield and trend data used in the analysis and provides motivation for the shifting-endpoint specification. Section 3 outlines the joint term structure model and its shifting-endpoint extension. Section 4 presents the results, and Section 5 concludes.

# 2 Data

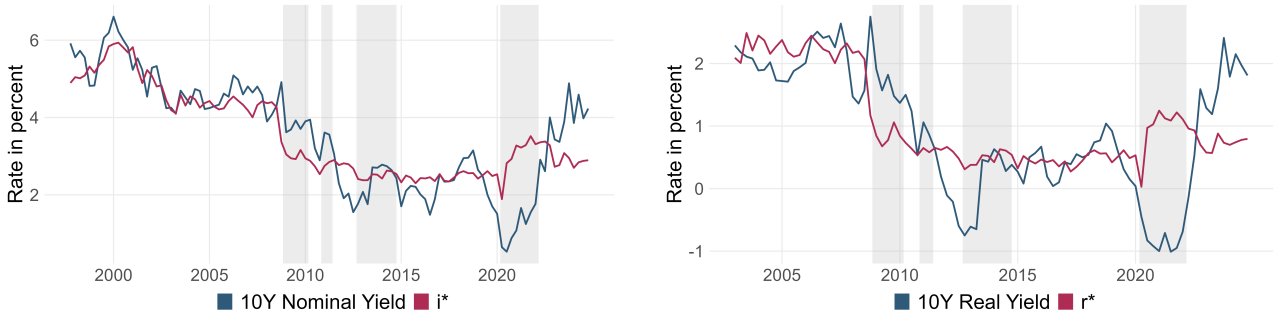
## 2.1 Bond Market Data

The real yield data are based on clean mid-market prices of Treasury Inflation-Protected Securities (TIPS), obtained from Refinitiv Eikon. We use individual TIPS prices rather than fitted yields in order to capture the liquidity premium on TIPS market prices. Over the sample period from August 1997 to December 2024, the dataset consists of $n_{\text{TIPS}} = 77$ individual TIPS with five- and ten-year maturities, consistent with the maturities considered by Andreasen et al. (2021). Nominal Treasury yields are obtained from Liu and Wu (2021), covering $n_y = 12$ maturities from three months to ten years. A complete overview of the characteristics of the nominal yield and individual TIPS data can be found in Appendix A.

## 2.2 Trend Data

To capture shifts in the long-run anchors of interest rates, we use external estimates of macroeconomic trends. We consider two macroeconomic trends, namely the equilibrium real interest rate $r_t^*$ and expected steady-state

inflation $\pi_t^*$. The equilibrium real interest rate reflects the real rate consistent with an economy operating at full employment and stable inflation, while the expected steady-state inflation reflects the perceived inflation target of the central bank over longer horizons. By the Fisher equation, their sum $i_t^* = r_t^* + \pi_t^*$ is the equilibrium nominal rate, which serves as the long-run anchor of the nominal yield curve. For the real equilibrium rate, we use the synthetic real-time estimates provided by Cho and Williams (2025), obtained using the model of Holston et al. (2017). To capture long-run inflation expectations, we utilize the perceived target rate (PTR) from the Federal Reserve Board's FRB/US model.




A: Ten-year nominal yield and $i_t^*$.

B: Ten-year real yield and $r_t^*$.

Figure 1: Long-term yields and their corresponding long-run trend anchors from August 1997 to December 2024. Panel A shows this comparison for nominal yields, while Panel B shows the same for real yields. Shaded areas indicate periods of quantitative easing.

Figure 1 shows that the ten-year nominal and real yields track their respective trend anchors, but vary considerably more in the short run. The shaded regions denote periods of quantitative easing, which coincide with larger temporary deviations of yields from their trend anchors. Taken together, these patterns suggest that long-term yields are shaped by slow-moving fundamental trends, while unconventional monetary policy and financial instability can temporarily distort this relationship. We aim to capture the co-movement directly by tying the long-run means of nominal and real yields to their respective trends.

# 3 Methodology

For this study, we base our analysis on the framework proposed in Andreasen et al. (2021). We first outline its shadow-rate specification as the baseline model, denoted as B-$G^L(5)$. We then introduce the shifting-endpoint specification, referred to as SE-$G^L(7)$, which extends the baseline model with time-varying macroeconomic trends similar to Bauer and Rudebusch (2020).

## 3.1 Baseline Model

The B-$G^L(5)$ model is a five-factor term structure model that jointly prices nominal yields and individual TIPS, while accounting for TIPS-specific liquidity premia. The state vector consists of five factors,

$$Z_t = \left[L_t^N,\ S_t,\ C_t,\ L_t^R,\ X_t^{liq}\right]',$$

where the first four factors are Nelson-Siegel yield curve components and the fifth captures bond-specific liquidity conditions. Following Christensen et al. (2010), the nominal and real yield curves contain distinct level factors, as the nominal level factor $L_t^N$ reflects both the equilibrium real rate and expected steady-state inflation, while the real level factor $L_t^R$ accounts only for the equilibrium real rate. In contrast, the slope and curvature factors, $S_t$ and $C_t$, are shared across yield curves through a scaling parameter $\alpha^R \in \mathbb{R}$, as they are strongly positively correlated. Lastly, the liquidity factor $X_t^{liq}$ captures the liquidity premium that investors require on individual TIPS, as the TIPS market is less actively traded than the nominal Treasury market (Andreasen et al., 2021).

The model specifies separate dynamics for the instantaneous nominal and frictionless real risk-free rates,

$$i_t = \max(L_t^N + S_t, 0), \tag{1}$$

$$r_t = L_t^R + \alpha^R S_t, \tag{2}$$

where the nominal short rate is constrained by the zero lower bound, motivated by the prolonged period of near-zero nominal rates in our sample. Real rates may take negative values and are therefore unconstrained. To accommodate the lower bound, we adopt the shadow-rate framework of Christensen and Rudebusch (2015).

### 3.1.1 Model Dynamics

Bond prices follow from the factor dynamics under the risk-neutral measure $\mathbb{Q}$, while the time-series behavior of the factors is governed by the physical measure $\mathbb{P}$. Under $\mathbb{Q}$, the state vector follows a multivariate Ornstein-Uhlenbeck process,

$$dZ_t = K_Z^{\mathbb{Q}} \left( \theta_Z^{\mathbb{Q}} - Z_t \right) dt + \Sigma \, dW_t^{\mathbb{Q}}, \tag{3}$$

where $W_t^{\mathbb{Q}}$ is a five-dimensional Brownian motion and $\Sigma$ is a volatility matrix, assumed diagonal as in Christensen et al. (2011). The $\mathbb{Q}$-dynamics impose the Nelson-Siegel factor loading structure through

$$K_Z^{\mathbb{Q}} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & \lambda & -\lambda & 0 & 0 \\ 0 & 0 & \lambda & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \kappa_{\text{liq}}^{\mathbb{Q}} \end{pmatrix}, \qquad \theta_Z^{\mathbb{Q}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \theta_{\text{liq}}^{\mathbb{Q}} \end{pmatrix},$$

where $\lambda$ is identical to the decay parameter in the original Nelson-Siegel specification, and governs the rate at which the slope and curvature loadings decay with maturity. The liquidity factor follows its own mean-reverting process under $\mathbb{Q}$, determined by the mean-reversion speed $\kappa_{\text{liq}}^{\mathbb{Q}}$ and long-term mean $\theta_{\text{liq}}^{\mathbb{Q}}$.

The relationship between the $\mathbb{Q}$- and $\mathbb{P}$-dynamics follows from Girsanov's theorem and is given by the change of measure

$$dW_t^{\mathbb{Q}} = dW_t^{\mathbb{P}} + \Gamma_t \, dt,$$

where $\Gamma_t$ represents the market price of risk. To preserve affine dynamics under the $\mathbb{P}$-measure, we adopt the essentially affine risk premium specification of Duffee (2002), $\Gamma_t = \gamma_0 + \gamma_1 Z_t$, where $\gamma_0 \in \mathbb{R}^5$ and $\gamma_1 \in \mathbb{R}^{5\times 5}$ are unrestricted. The flexibility of $\Gamma_t$ allows the $\mathbb{P}$-dynamics to be specified freely while preserving the dynamics under $\mathbb{Q}$. Consistent with Andreasen et al. (2021), the state vector under $\mathbb{P}$ follows

$$dZ_t = K_Z^{\mathbb{P}} \left( \theta_Z^{\mathbb{P}} - Z_t \right) dt + \Sigma \, dW_t^{\mathbb{P}}, \tag{4}$$

where the mean-reversion matrix $K_Z^{\mathbb{P}}$ is restricted to be diagonal and the unconditional means in $\theta_Z^{\mathbb{P}}$ are treated as constant, implying stationary factor dynamics. The impact of the diagonal restriction is limited, as it results in similar model fit and $\mathbb{P}$-dynamics (Christensen et al., 2025).

## 3.2 Shifting-Endpoint Extension

We extend the baseline model through a trend-cycle decomposition of the short rates, anchoring the long-run means of the level factors to the equilibrium real rate $r_t^*$ and long-run inflation expectations $\pi_t^*$. Although the short rate depends on both the level and slope factors, we assume that its time-varying long-run component is fully captured by the level factor. This assumption is consistent with the shifting-endpoint literature, where most forecasting gains arise from the level factor (Opschoor & Van der Wel, 2025; Van Dijk et al., 2014).

### 3.2.1 Trend-Cycle Decomposition of the Short Rates

Following Bauer and Rudebusch (2020), the trend-cycle decomposition adopts a standard unobserved components framework as in Watson (1986). Specifically, the nominal and real short rates in equations (1) and (2) are decomposed as

$$i_t = i_t^* + i_t^c, \qquad i_t^* = i_{t-1}^* + \eta_t^{i^*}, \tag{5}$$

$$r_t = r_t^* + r_t^c, \qquad r_t^* = r_{t-1}^* + \eta_t^{r^*}, \tag{6}$$

where the equilibrium rates $i_t^*$ and $r_t^*$ are the non-stationary trend components of the short rates, specified to follow independent random walk processes. The cyclical components $i_t^c$ and $r_t^c$ are stationary, zero-mean processes that capture short-run fluctuations around these trends. Their dynamics are given by a VAR(1) process implied by the discretization of equation (4).

### 3.2.2 Non-Stationary Level Factors

We implement the trend-cycle decomposition by allowing the long-run means of the level factors to vary over time, replacing the fixed parameters $\theta_1^{\mathbb{P}}$ and $\theta_4^{\mathbb{P}}$ in equation (4) with time-varying parameters $\theta_{1,t}^{\mathbb{P}}$ and $\theta_{4,t}^{\mathbb{P}}$. Combining the short rate definitions in equations (1) and (2) with the decomposition in equations (5) and (6)

reveals how the trend and cyclical components arise from the factors through their long-run means. In terms of the nominal shadow rate and the real short rate, this gives

$$i_t = L_t^N + S_t \quad = \underbrace{\theta_{1,t}^{\mathbb{P}} + \theta_2^{\mathbb{P}}}_{i_t^*} + \underbrace{(L_t^N - \theta_{1,t}^{\mathbb{P}}) + (S_t - \theta_2^{\mathbb{P}})}_{i_t^c}, \tag{7}$$

$$r_t = L_t^R + \alpha^R S_t = \underbrace{\theta_{4,t}^{\mathbb{P}} + \alpha^R \theta_2^{\mathbb{P}}}_{r_t^*} + \underbrace{(L_t^R - \theta_{4,t}^{\mathbb{P}}) + \alpha^R (S_t - \theta_2^{\mathbb{P}})}_{r_t^c}, \tag{8}$$

where $\theta_2^{\mathbb{P}}$ denotes the fixed long-run mean of the slope factor. The trend components combine the time-varying level means with the long-run contribution of the slope factor, defining the shifting endpoints toward which the short rates converge at long horizons. The cyclical components correspond to demeaned factor paths. Rearranging the trend components yields

$$\theta_{1,t}^{\mathbb{P}} = i_t^* - \theta_2^{\mathbb{P}} = \pi_t^* + r_t^* - \theta_2^{\mathbb{P}}, \qquad \theta_{4,t}^{\mathbb{P}} = r_t^* - \alpha^R \theta_2^{\mathbb{P}}, \tag{9}$$

where the second equality for $\theta_{1,t}^{\mathbb{P}}$ follows from the Fisher equation. Since the trends $\pi_t^*$ and $r_t^*$ follow random walks, they are difficult to identify from yields alone. We therefore use the observable proxies described in Section 2.2 to identify the underlying trends, treating them as imperfect measurements. This approach allows the proxies to provide additional information about the trends without imposing exact equality, with the full specification outlined in Appendix C.4.3.

## 3.3 Pricing Equations

The arbitrage-free pricing structure is shared between both model specifications and follows from the factor dynamics under $\mathbb{Q}$. Although the SE-$G^L(7)$ model decomposes the nominal and real level factors into trend and cyclical components, these components enter bond prices through their sum, leaving the arbitrage-free pricing relations unchanged.

### 3.3.1 Nominal Yields

Within the shadow-rate framework of Christensen and Rudebusch (2015), the nominal yield at maturity $\tau$ is obtained by integrating over a nonnegative forward rate,

$$y_t^N(\tau) = \frac{1}{\tau} \int_0^\tau \underline{f}_t(s)\, ds, \qquad \underline{f}_t(\tau) = f_t(\tau)\, \Phi\left(\frac{f_t(\tau)}{\omega(\tau)}\right) + \omega(\tau)\, \phi\left(\frac{f_t(\tau)}{\omega(\tau)}\right), \tag{10}$$

where $\Phi(\cdot)$ and $\phi(\cdot)$ denote the standard normal CDF and PDF, $f_t(\tau)$ is the unconstrained forward rate following the Nelson-Siegel structure, and $\omega(\tau)$ is the maturity-dependent conditional standard deviation implied by the $\mathbb{Q}$-dynamics. Closed-form expressions for $f_t(\tau)$ and $\omega(\tau)$ are reported in Appendices B.1 and B.2. We evaluate the integral numerically using a Riemann sum with step size $\Delta s = 0.01$ years. When the unconstrained forward rate remains strictly positive across all maturities, the lower-bound adjustment becomes negligible and nominal yields reduce to the affine specification.

### 3.3.2 Real Yields

Real yields follow the Nelson-Siegel factor-loading structure with a bond-specific liquidity adjustment. At time $t$, the yield on the $i$th TIPS with issuance date $t_0^i$ and remaining maturity $\tau$, is

$$\begin{aligned} y_t^{R,i}(t_0^i, t, \tau) = & \underbrace{L_t^R + \alpha^R \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau}\right) S_t + \alpha^R \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau}\right) C_t}_{\text{frictionless loadings}} \\ & + \underbrace{\beta^i \left(\frac{1 - e^{-\kappa_{liq}^Q \tau}}{\kappa_{liq}^Q \tau} - e^{-\lambda^{L,i}(t - t_0^i)} \frac{1 - e^{-(\kappa_{liq}^Q + \lambda^{L,i})\tau}}{(\kappa_{liq}^Q + \lambda^{L,i})\tau}\right) X_t^{liq}}_{\text{liquidity adjustment}} \quad - \quad \underbrace{\frac{A^{R,i}(t_0^i, t, \tau)}{\tau}}_{\text{deterministic adjustment}}, \end{aligned} \tag{11}$$

where the first component captures the frictionless real yield implied by the Nelson-Siegel factors, the second component reflects exposure to the liquidity factor $X_t^{liq}$, and the final term is the adjustment ensuring absence of arbitrage. As the model is estimated directly on individual TIPS prices rather than fitted yields, we use equation (11) to construct the corresponding clean bond prices. Further details on the construction of TIPS prices and the deterministic adjustment term are provided in Appendix B.3 and Appendix B.4, respectively.

### 3.4 Estimation and Evaluation

Both models are estimated by quasi-maximum likelihood using the extended Kalman filter, which is required because the measurement equations are nonlinear in the state variables. First, individual TIPS need to be discounted, so the clean price in Appendix B.3 is a nonlinear function of the state vector. Second, the shadow-rate framework introduces an additional nonlinearity in the nominal short rate, defined in equation (1), to account for the zero lower bound. Further details on the estimation procedure are provided in Appendix C.

We evaluate in-sample fit using pricing errors and information criteria, and additionally examine the trend-cycle decomposition of the level factors under the shifting-endpoint specification. Out-of-sample, we forecast nominal and real ten-year yields at a monthly frequency for horizons ranging from 6 to 24 months, with annual re-estimation. Forecasts are constructed using an expanding window beginning in January 2014. A random walk serves as an additional benchmark, which is difficult to outperform given the strong persistence of interest rates (Duffee, 2013). Significance is tested with a Diebold-Mariano test using the small-sample correction of Harvey et al. (1997) and Newey-West standard errors for overlapping horizons. The appendix reports a subsample analysis using the sample of Andreasen et al. (2021), residual diagnostics, and additional forecast results.

## 4 Results

### 4.1 In-Sample Analysis

The pricing errors for nominal yields and TIPS are reported in Table 1, measured in terms of the mean error and the root mean squared error (RMSE). The SE-$G^L(7)$ model consistently outperforms the B-$G^L(5)$ model in overall RMSEs by approximately 0.3 basis points. For nominal yields, the pricing errors are approximately 1.4 basis points higher than for TIPS. The model fit measures also favor the SE-$G^L(7)$ model, with a higher log-likelihood and lower AIC and BIC, suggesting that incorporating shifting endpoints improves overall model fit. The estimated factor paths are very similar across specifications and are reported in Appendix D.1, together with the corresponding model parameters in Appendix D.2. Since the error terms are not normally distributed, shown by the residual analysis in Appendix D.3, we cannot formally test the improvement with a likelihood ratio test, as the critical values are invalid (White, 1982). The improved performance of the SE-$G^L(7)$ model also holds in the subsample, as shown in Appendix D.4, although it is slightly smaller.

Table 1: Pricing Errors: Full Sample

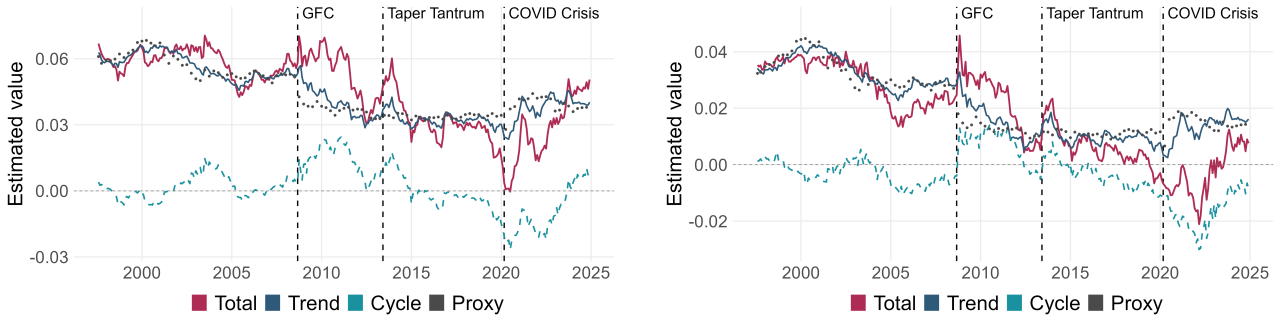
| | B-$G^L(5)$ | | SE-$G^L(7)$ | |
|---|---|---|---|---|
| Maturity in months | Mean | RMSE | Mean | RMSE |
| Panel A: Nominal yields | | | | |
| 60 | −1.87 | 4.70 | −1.98 | 4.61 |
| 120 | 1.79 | 6.42 | 2.17 | 6.07 |
| All Nominal | −0.53 | 5.92 | −0.41 | 5.59 |
| Panel B: TIPS (averaged by maturity) | | | | |
| 60 | −0.66 | 6.27 | −0.77 | 5.87 |
| 120 | −0.10 | 4.07 | −0.06 | 3.81 |
| All TIPS | −0.20 | 4.53 | −0.19 | 4.25 |
| Log-likelihood | 50820.32 | | 51136.98 | |
| AIC | −101290.64 | | −101919.96 | |
| BIC | −100626.33 | | −101248.06 | |

This table reports mean pricing errors (Mean) and root mean squared errors (RMSE) in basis points for nominal yields and TIPS from August 1997 to December 2024. Results are reported for the B-$G^L(5)$ and SE-$G^L(7)$ models, both estimated with diagonal $K_Z^{\mathbb{P}}$ and $\Sigma$ matrices. The pricing errors are defined as observed minus model-implied yields. For TIPS, market prices are converted to yields to maturity and averaged by maturity group. The final rows report the log-likelihood, Akaike information criterion (AIC), and Bayesian information criterion (BIC).

#### 4.1.1 Level Factor Decomposition Analysis

The richer dynamics of the SE-$G^L(7)$ model allow us to decompose the level factors into a trend and a cyclical component. Figure 2 presents the results for the nominal and real level factors, together with the corresponding

trend proxies adjusted for the unconditional mean of the slope factor. The level factors closely track their trend components, while the cyclical components fluctuate around zero. The descriptive statistics in Appendix D.5 confirm this pattern. The cyclical components have sample means close to zero and autocorrelations that decline with the lag length, in contrast to the highly persistent trend components. Formal evidence of stationarity is harder to establish, as unit root tests have low power in this relatively small sample. The Augmented Dickey-Fuller test fails to reject the unit root null for either cyclical component.




A: Nominal level factor: trend and cyclical components.

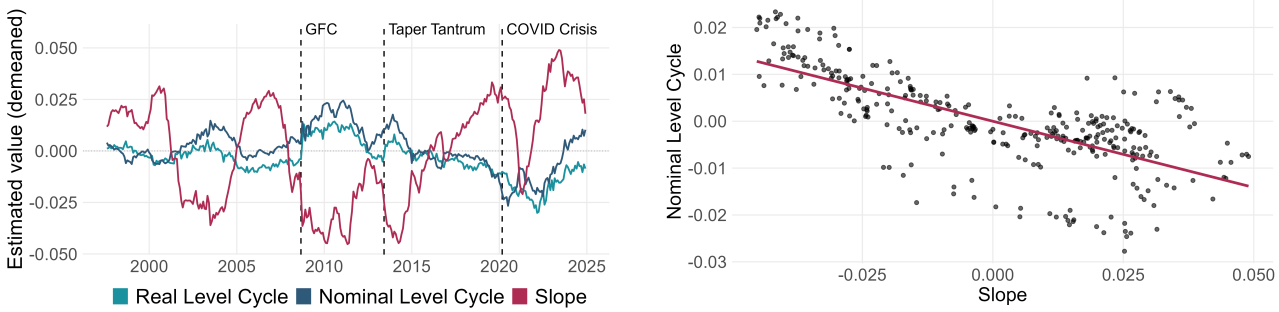
B: Real level factor: trend and cyclical components.

Figure 2: Trend-cycle decomposition of the nominal and real level factors in the SE-$G^L(7)$ model from August 1997 to December 2024. Dots denote the proxy series adjusted for the unconditional mean of the slope factor.

Since the level factor shifts all yields equally, the cyclical component indicates whether yields are temporarily above or below their long-run anchor. Positive cycles indicate elevated yields during periods of heightened uncertainty, such as after the global financial crisis (GFC) and during the taper tantrum. In contrast, the COVID crisis is characterized by a negative cycle, as large-scale asset purchases pushed yields below the trend. These episodes illustrate that the cyclical component captures transitory deviations from trend that can be traced to macroeconomic and monetary policy events.

### 4.1.2 Relationship between the Level and Slope Factor

The trend-cycle decomposition also reveals a negative relationship between the cyclical level component and the slope factor, illustrated in Figure 3. The nominal and real level cycles display strong negative co-movement with the slope factor, with correlations of $-0.70$ and $-0.64$, respectively. This finding is consistent with Diebold et al. (2006), who show that macroeconomic shocks affect the level and slope factors in opposite directions. This negative co-movement also provides insight into yield curve movements. The slope factor varies more than the cyclical level component, so its effect dominates at shorter maturities. At longer maturities, yields are primarily driven by the level factor as the loading on the slope factor declines with maturity. An increase in the cyclical level component therefore raises longer-term yields, while short-term yields decline under the stronger influence of the slope factor. Periods of heightened economic uncertainty, such as the global financial crisis and the taper tantrum, are thus associated with a temporary steepening of the yield curve.




A: Demeaned cyclical level components and slope factor.

B: Scatter plot of nominal level cycle and slope factor.

Figure 3: The relationship between the cyclical level components and the slope factor from August 1997 to December 2024. The left panel shows their movements over time, while the right panel shows their correlation.

## 4.2 Out-Of-Sample Analysis

Table 2 reports the root mean squared forecast errors (RMSFEs) for the ten-year nominal and real yield across four forecast horizons. The SE-$G^L(7)$ model achieves the lowest forecast errors in almost all cases. The only exception is the ten-year real yield at the six-month horizon, where the random walk performs slightly better

(51.14 compared to 55.12 basis points). The differences across specifications become more pronounced at longer horizons, with larger gains from shifting endpoints for real yields than for nominal yields. The improvements of the SE-$G^L(7)$ model over the B-$G^L(5)$ model are statistically significant at the conventional 5% level for all nominal and real yield forecasts across all horizons, based on the adjusted Diebold-Mariano test. These results are robust to forecasting five-year yields instead. Moreover, excluding the COVID period from March 2020 to March 2021 yields similar results, ensuring that they are not shaped by the pandemic. In both cases, the SE-$G^L(7)$ model remains the preferred specification, with full results in Appendices E.1 and E.2, respectively.

Table 2: Out-of-sample Performance for Ten-year Yields

| | Forecast Horizon (in months) | | | |
|---|---|---|---|---|
| | 6 (T = 126) | 12 (T = 120) | 18 (T = 114) | 24 (T = 108) |
| Panel A: Nominal ten-year yield | | | | |
| B-$G^L(5)$ | 61.02 | 90.84 | 117.61 | 146.38 |
| SE-$G^L(7)$ | **57.96**$^*$ | **82.92**$^*$ | **103.63**$^*$ | **127.17**$^*$ |
| RW | 61.27 | 92.30 | 120.69 | 151.30 |
| Panel B: Real ten-year yield | | | | |
| B-$G^L(5)$ | 61.09 | 87.97 | 116.23 | 141.19 |
| SE-$G^L(7)$ | 55.12$^*$ | **72.70**$^*$ | **89.96**$^*$ | **107.34**$^*$ |
| RW | **51.14** | 81.65 | 111.01 | 137.00 |

This table reports root mean squared forecast errors (RMSFEs) in basis points for forecasts of nominal and real ten-year yields from January 2014 to December 2024. The models are B-$G^L(5)$, SE-$G^L(7)$, and a random walk benchmark (RW), evaluated at horizons $h = 6, 12, 18$, and 24 months. The lowest RMSFE within each panel is highlighted in bold. An asterisk (*) indicates that SE-$G^L(7)$ significantly outperforms B-$G^L(5)$ at the 5% significance level, based on a one-sided Diebold-Mariano test with the Harvey et al. (1997) small-sample correction and Newey-West standard errors with $h - 1$ lags.

As the forecast horizon lengthens, the SE-$G^L(7)$ model separates more clearly from the benchmarks. At short horizons, model differences are limited, with forecast errors varying by at most 10 basis points at $h = 6$ months. However, as the forecast horizon increases, differences reach up to 34 basis points at $h = 24$ months. This highlights the role of shifting endpoints, as the increasing importance of mean reversion at longer horizons leads fixed-endpoint models to revert toward a constant long-run level that may be misspecified.

# 5 Conclusion

In this paper, we examine whether incorporating shifting endpoints improves in-sample fit and leads to better out-of-sample performance. We do so by extending the five-factor arbitrage-free Nelson-Siegel model of Andreasen et al. (2021) with shifting endpoints linked to observable macroeconomic trends. This extension allows the long-run means of the nominal and real level factors to be driven separately by the equilibrium real rate and trend inflation. Separately identifying the two trend components tells policymakers and investors whether a move in the yield curve reflects a shift in the equilibrium real rate, a change in long-run inflation expectations, or a temporary deviation from trend.

Incorporating shifting endpoints improves both model fit and forecast accuracy. The shifting-endpoint model attains the highest log-likelihood and the lowest AIC and BIC values in comparison to the fixed-endpoint baseline, with overall RMSEs of 5.59 basis points for nominal yields and 4.25 basis points for TIPS over the full sample. The trend-cycle decomposition separates slowly evolving trends and mean-reverting cycles, revealing a negative relationship between the cyclical level component and the slope factor. Out-of-sample, the shifting-endpoint specification achieves the lowest forecasting errors across nearly all horizons for both nominal and real yields, significantly outperforming its fixed-endpoint counterpart at every horizon. The improvements become more pronounced as the forecast horizon increases, reflecting the growing importance of mean reversion toward a correctly specified long-run anchor at longer horizons.

The results also point to two directions for future research. First, a Bayesian approach with fully latent trends, as in Bauer and Rudebusch (2020), would produce purely market-based estimates of the equilibrium real rate and long-run inflation expectations from market prices alone, without relying on external proxy series. Second, incorporating observable macroeconomic variables, as in Diebold et al. (2006), could provide new insights on the common underlying forces driving the cyclical co-movement of the level and slope factors.

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# Appendix A: Bond Data Overview

## A.1 TIPS Characteristics

Table 3: Characteristics of 5-Year and 10-Year TIPS

| 5-Year TIPS | | | | 10-Year TIPS | | | |
|---|---|---|---|---|---|---|---|
| CUSIP | Coupon rate (%) | Auction date | Maturity date | CUSIP | Coupon rate (%) | Auction date | Maturity date |
| 9128273A8 | 3.625 | 1997-07-09 | 2002-07-15 | 9128272M3 | 3.375 | 1997-01-29 | 2007-01-15 |
| 912828CZ1 | 0.875 | 2004-10-26 | 2010-04-15 | 9128273T7 | 3.625 | 1998-01-08 | 2008-01-15 |
| 912828FB1 | 2.375 | 2006-04-25 | 2011-04-15 | 9128274Y5 | 3.875 | 1999-01-06 | 2009-01-15 |
| 912828GN4 | 2.000 | 2007-04-24 | 2012-04-15 | 9128275W8 | 4.250 | 2000-01-12 | 2010-01-15 |
| 912828HW3 | 0.625 | 2008-04-22 | 2013-04-15 | 9128276R8 | 3.500 | 2001-01-10 | 2011-01-15 |
| 912828KM1 | 1.250 | 2009-04-23 | 2014-04-15 | 9128277J5 | 3.375 | 2002-01-09 | 2012-01-15 |
| 912828MY3 | 0.500 | 2010-04-26 | 2015-04-15 | 912828AF7 | 3.000 | 2002-07-10 | 2012-07-15 |
| 912828QD5 | 0.125 | 2011-04-21 | 2016-04-15 | 912828BD1 | 1.875 | 2003-07-09 | 2013-07-15 |
| 912828SQ4 | 0.125 | 2012-04-19 | 2017-04-15 | 912828BW9 | 2.000 | 2004-01-08 | 2014-01-15 |
| 912828UX6 | 0.125 | 2013-04-18 | 2018-04-15 | 912828CP3 | 2.000 | 2004-07-08 | 2014-07-15 |
| 912828C99 | 0.125 | 2014-04-17 | 2019-04-15 | 912828DH0 | 1.625 | 2005-01-13 | 2015-01-15 |
| 912828K33 | 0.125 | 2015-04-23 | 2020-04-15 | 912828EA4 | 1.875 | 2005-07-14 | 2015-07-15 |
| 912828Q60 | 0.125 | 2016-04-21 | 2021-04-15 | 912828ET3 | 2.000 | 2006-01-12 | 2016-01-15 |
| 912828X39 | 0.125 | 2017-04-20 | 2022-04-15 | 912828FL9 | 2.500 | 2006-07-13 | 2016-07-15 |
| 9128284H0 | 0.625 | 2018-04-19 | 2023-04-15 | 912828GD6 | 2.375 | 2007-01-11 | 2017-01-15 |
| 9128286N5 | 0.500 | 2019-04-18 | 2024-04-15 | 912828GX2 | 2.625 | 2007-07-12 | 2017-07-15 |
| 912828YL8 | 0.125 | 2019-10-17 | 2024-10-15 | 912828HN3 | 1.625 | 2008-01-10 | 2018-01-15 |
| 912828ZJ2 | 0.125 | 2020-04-23 | 2025-04-15 | 912828JE1 | 1.375 | 2008-07-10 | 2018-07-15 |
| 91282CAQ4 | 0.125 | 2020-10-22 | 2025-10-15 | 912828JX9 | 2.125 | 2009-01-06 | 2019-01-15 |
| 91282CCA7 | 0.125 | 2021-04-22 | 2026-04-15 | 912828LA6 | 1.875 | 2009-07-06 | 2019-07-15 |
| 91282CDC2 | 0.125 | 2021-10-21 | 2026-10-15 | 912828MF4 | 1.375 | 2010-01-11 | 2020-01-15 |
| 91282CEJ6 | 0.125 | 2022-04-21 | 2027-04-15 | 912828NM8 | 1.250 | 2010-07-08 | 2020-07-15 |
| 91282CFR7 | 1.625 | 2022-10-20 | 2027-10-15 | 912828PP9 | 1.125 | 2011-01-20 | 2021-01-15 |
| 91282CGW5 | 1.250 | 2023-04-20 | 2028-04-15 | 912828QV5 | 0.625 | 2011-07-21 | 2021-07-15 |
| 91282CJH5 | 2.375 | 2023-10-19 | 2028-10-15 | 912828SA9 | 0.125 | 2012-01-19 | 2022-01-15 |
| 91282CKL4 | 2.125 | 2024-04-18 | 2029-04-15 | 912828TE0 | 0.125 | 2012-07-19 | 2022-07-15 |
| 91282CLV1 | 1.625 | 2024-10-24 | 2029-10-15 | 912828UH1 | 0.125 | 2013-01-24 | 2023-01-15 |
| | | | | 912828VM9 | 0.375 | 2013-07-18 | 2023-07-15 |
| | | | | 912828B25 | 0.625 | 2014-01-23 | 2024-01-15 |
| | | | | 912828WU0 | 0.125 | 2014-07-24 | 2024-07-15 |
| | | | | 912828H45 | 0.250 | 2015-01-22 | 2025-01-15 |
| | | | | 912828XL9 | 0.375 | 2015-07-23 | 2025-07-15 |
| | | | | 912828N71 | 0.625 | 2016-01-21 | 2026-01-15 |
| | | | | 912828S50 | 0.125 | 2016-07-21 | 2026-07-15 |
| | | | | 912828V49 | 0.375 | 2017-01-19 | 2027-01-15 |
| | | | | 9128282L3 | 0.375 | 2017-07-20 | 2027-07-15 |
| | | | | 9128283R9 | 0.500 | 2018-01-18 | 2028-01-15 |
| | | | | 912828Y38 | 0.750 | 2018-07-19 | 2028-07-15 |
| | | | | 9128285W6 | 0.875 | 2019-01-17 | 2029-01-15 |
| | | | | 9128287D6 | 0.250 | 2019-07-18 | 2029-07-15 |
| | | | | 912828Z37 | 0.125 | 2020-01-23 | 2030-01-15 |
| | | | | 912828ZZ6 | 0.125 | 2020-07-23 | 2030-07-15 |
| | | | | 91282CBF7 | 0.125 | 2021-01-21 | 2031-01-15 |
| | | | | 91282CCM1 | 0.125 | 2021-07-22 | 2031-07-15 |
| | | | | 91282CDX6 | 0.125 | 2022-01-20 | 2032-01-15 |
| | | | | 91282CEZ0 | 0.625 | 2022-07-21 | 2032-07-15 |
| | | | | 91282CGK1 | 1.125 | 2023-01-19 | 2033-01-15 |
| | | | | 91282CHP9 | 1.375 | 2023-07-20 | 2033-07-15 |
| | | | | 91282CJY8 | 1.750 | 2024-01-18 | 2034-01-15 |
| | | | | 91282CLE9 | 1.875 | 2024-07-18 | 2034-07-15 |

This table reports CUSIP, coupon rate, auction date, and maturity date for U.S. 5-year and 10-year Treasury Inflation-Protected Securities (TIPS) used in this study.

## A.2 Nominal Yields Characteristics

Table 4: Descriptive Statistics of U.S. Nominal Treasury Yields

| Maturity | Mean | StdDev | Min | Max | $\rho_1$ | $\rho_{12}$ | $\rho_{24}$ |
|---|---|---|---|---|---|---|---|
| 3 | 2.099 | 2.045 | 0.009 | 6.280 | 0.995 | 0.737 | 0.281 |
| 6 | 2.186 | 2.045 | 0.028 | 6.275 | 0.995 | 0.743 | 0.295 |
| 12 | 2.303 | 2.017 | 0.064 | 6.561 | 0.994 | 0.754 | 0.324 |
| 24 | 2.445 | 1.905 | 0.123 | 6.622 | 0.991 | 0.777 | 0.405 |
| 36 | 2.616 | 1.799 | 0.123 | 6.587 | 0.989 | 0.795 | 0.472 |
| 48 | 2.800 | 1.706 | 0.172 | 6.604 | 0.987 | 0.803 | 0.515 |
| 60 | 2.962 | 1.627 | 0.231 | 6.672 | 0.986 | 0.806 | 0.550 |
| 72 | 3.117 | 1.558 | 0.313 | 6.644 | 0.985 | 0.810 | 0.576 |
| 84 | 3.248 | 1.508 | 0.382 | 6.637 | 0.984 | 0.815 | 0.607 |
| 96 | 3.362 | 1.473 | 0.445 | 6.622 | 0.983 | 0.821 | 0.633 |
| 108 | 3.465 | 1.449 | 0.488 | 6.608 | 0.983 | 0.825 | 0.650 |
| 120 | 3.570 | 1.441 | 0.530 | 6.608 | 0.983 | 0.832 | 0.673 |

This table reports summary statistics (in percent) and autocorrelations at lags 1, 12, and 24 for each maturity. The sample consists of monthly observations from August 1997 to December 2024, and maturities are denoted in months.

# Appendix B: Derivation of the Pricing Equations

This appendix collects the expressions that underlie the nominal yield and TIPS pricing equations in Section 3.3. All expressions follow from the $\mathbb{Q}$-dynamics specified in equation (3) under the assumption of a diagonal volatility matrix $\Sigma$. The adjustment term for nominal yields is derived in Christensen et al. (2011). The forward-rate adjustment term and the conditional variance term used in the shadow-rate specification follow Christensen and Rudebusch (2015), adapted to the case of a diagonal volatility matrix. The real bond adjustment terms follow from the online appendix of Andreasen et al. (2021).

## B.1 Unconstrained Forward Rate

Nominal yields follow from the nonnegative forward rate in equation (10), which adjusts the unconstrained forward rate for the lower bound. The unconstrained forward rate is affine in the state vector and given by

$$f_t(\tau) = L_t^N + e^{-\lambda\tau} S_t + \lambda\tau e^{-\lambda\tau} C_t + A^f(\tau), \tag{12}$$

where $\lambda$ is the decay parameter governing the slope and curvature loadings. Only the nominal level, slope, and curvature factors enter, as the real level and liquidity factors do not affect nominal yields. The adjustment term $A^f(\tau)$ ensures absence of arbitrage and is implied by the risk neutral dynamics. It is obtained by differentiating the nominal log bond price adjustment with respect to maturity,

$$A^f(\tau) = -\frac{\partial A(\tau)}{\partial \tau}, \tag{13}$$

where $A(\tau) \equiv -\tau \cdot A^N(\tau)/\tau$. Under the diagonal $\Sigma$ restriction, the adjustment term is given by

$$A^f(\tau) = -\frac{1}{2}\sigma_{11}^2\tau^2 - \frac{1}{2}\sigma_{22}^2\left(\frac{1-e^{-\lambda\tau}}{\lambda}\right)^2 - \frac{1}{2}\sigma_{33}^2\left(\tau e^{-\lambda\tau} - \frac{1-e^{-\lambda\tau}}{\lambda}\right)^2. \tag{14}$$

## B.2 Conditional Variance Term

The nonnegative forward rate in equation (10) depends on the conditional variance term $\omega(\tau)^2$, which scales the lower-bound adjustment applied to the unconstrained forward rate. Under the diagonal $\Sigma$ restriction, it is given by

$$\omega(\tau)^2 = \sigma_{11}^2\tau + \sigma_{22}^2\frac{1-e^{-2\lambda\tau}}{2\lambda} + \sigma_{33}^2\left(\frac{1-e^{-2\lambda\tau}}{4\lambda} - \frac{\tau e^{-2\lambda\tau}}{2} - \frac{\lambda\tau^2 e^{-2\lambda\tau}}{2}\right). \tag{15}$$

### B.3 TIPS Clean Bond Price

The clean bond price of the $i$th TIPS with issuance date $t_0^i$, pricing date $t$, and maturity $T$ is obtained by discounting each coupon payment using the liquidity-adjusted real yield in equation (11),

$$\begin{aligned} P^{\text{TIPS,i}}\left(t_0^i, t, T\right) = C\left(t_1 - t\right)\exp\left\{-\left(t_1 - t\right) y_t^{R,i}\left(t_0^i, t, t_1\right)\right\} \\ + \sum_{k=2}^{n} \frac{C}{2}\exp\left\{-\left(t_k - t\right) y_t^{R,i}\left(t_0^i, t, t_k\right)\right\} \\ + \exp\left\{-(T-t) y_t^{R,i}\left(t_0^i, t, T\right)\right\}, \end{aligned} \tag{16}$$

where $C$ denotes the coupon rate and $t_k$ are the remaining payment dates. The model does not incorporate the deflation option value embedded in TIPS, as the option is rarely in the money and has only a negligible effect on TIPS prices (Andreasen et al., 2021).

### B.4 Real Yield Adjustment Term

The general frictionless real yield-adjustment term $A^R(\tau)/\tau$ depends on the volatilities of the real level factor, the shared slope and curvature factors scaled by $\alpha^R$, and the decay parameter $\lambda$. Under the diagonal $\Sigma$ restriction, its closed-form expression is

$$\begin{aligned} \frac{A^R(\tau)}{\tau} = \sigma_{44}^2\frac{\tau}{6} + \sigma_{22}^2(\alpha^R)^2\left[\frac{1}{2\lambda^2} - \frac{1}{\lambda^3}\frac{1-e^{-\lambda\tau}}{\tau} + \frac{1}{4\lambda^3}\frac{1-e^{-2\lambda\tau}}{\tau}\right] \\ + \sigma_{33}^2(\alpha^R)^2\left[\frac{1}{2\lambda^2} + \frac{1}{\lambda^2}e^{-\lambda\tau} - \frac{1}{4\lambda}\tau e^{-2\lambda\tau} - \frac{3}{4\lambda^2}e^{-2\lambda\tau} + \frac{5}{8\lambda^3}\frac{1-e^{-2\lambda\tau}}{\tau} - \frac{2}{\lambda^3}\frac{1-e^{-\lambda\tau}}{\tau}\right]. \end{aligned} \tag{17}$$

For individual TIPS, the bond-specific adjustment $A^{R,i}(t_0^i, t, \tau)/\tau$ in equation (11) additionally includes terms that capture the effect of the liquidity factor drift and its convexity correction, which depend on $\beta^i$, $\lambda_{L,i}$, $\kappa_{\text{liq}}^{\mathbb{Q}}$, $\theta_{\text{liq}}^{\mathbb{Q}}$, and $\sigma_{55}$. They follow from the system of ordinary differential equations in Andreasen et al. (2021), Online Appendix A, equations 6 and 7, with a diagonal covariance matrix $\Sigma$.

## Appendix C: Extended Kalman Filter Estimation

Both models are estimated by quasi-maximum likelihood using the extended Kalman filter (EKF), following Christensen and Rudebusch (2015) and the supplementary appendix of Andreasen et al. (2021). The EKF is needed because the measurement equations are nonlinear in the state variables. We first cast the B-$G^L(5)$ model in state-space form, outline the estimation procedure, and then describe the modifications required for the shifting-endpoint extension.

### C.1 Discretization

The transition equation is obtained by exact discretization of the continuous-time $\mathbb{P}$-dynamics in equation (4). Under the diagonal specification of $K_Z^{\mathbb{P}}$, this yields a VAR(1) process for the state vector $Z_t$,

$$Z_t = \Phi_0 + \Phi_1 Z_{t-1} + \xi_t, \qquad \xi_t \sim N(0,\, Q), \tag{18}$$

where $\Phi_0 = (I - \Phi_1)\theta_Z^{\mathbb{P}}$, $\Phi_1 = \exp(-K_Z^{\mathbb{P}}\Delta t)$ is diagonal with entries $e^{-\kappa_{jj}\Delta t}$, and $\Delta t = 1/12$ is the monthly time step. The state noise covariance matrix follows from the conditional variance of the Ornstein-Uhlenbeck process. Under the diagonal restrictions, $Q$ is diagonal with elements

$$Q_{jj} = \frac{\sigma_{jj}^2\left(1 - e^{-2\kappa_{jj}\Delta t}\right)}{2\kappa_{jj}}, \qquad j = 1, \ldots, 5,$$

for which closed-form solutions are provided by Fisher and Gilles (1996). The transition and measurement errors, $\xi_t$ and $\varepsilon_t$, are assumed to be mutually independent and orthogonal to the initial state,

$$\begin{pmatrix}\xi_t \\ \varepsilon_t\end{pmatrix} \sim N\left(\begin{pmatrix}0 \\ 0\end{pmatrix}, \begin{pmatrix}Q & 0 \\ 0 & H_t\end{pmatrix}\right),$$

where $H_t$ is the measurement error covariance matrix defined in the next section.

## C.2 Measurement Equations

The observation vector stacks the nominal yields and the duration-scaled TIPS prices. Each element maps the state vector into an observable through a maturity-specific intercept and a loading vector, denoted $A$ and $B$ throughout. The measurement error covariance matrix $H_t$ collects the corresponding pricing errors and is specified as diagonal,

$$H_t = \mathrm{diag}\Big(\underbrace{\sigma_y^2, \ldots, \sigma_y^2}_{n_y}, \underbrace{\sigma_{\mathrm{TIPS}}^2, \ldots, \sigma_{\mathrm{TIPS}}^2}_{n_t^{\mathrm{TIPS}}}\Big),$$

where $n_y = 12$ is the fixed number of nominal yield maturities and $n_t^{\mathrm{TIPS}}$ is the number of TIPS in the cross-section at time $t$, which varies as bonds are issued and mature. Following Joslin et al. (2011), all nominal yields have independent pricing errors with a common variance $\sigma_y^2$. The common pricing error variances $\sigma_y^2$ and $\sigma_{\mathrm{TIPS}}^2$ are estimated jointly with the model parameters.

### C.2.1 Nominal Yields

The measurement equation for the nominal yield at maturity $\tau_k$ follows from the constrained forward rate in equation (10),

$$y_t^N(\tau_k) = z^N(Z_t;\, \tau_k) + \varepsilon_t^{y,k}, \qquad z^N(Z_t;\, \tau_k) = \frac{1}{\tau_k}\int_0^{\tau_k} \underline{f}_t(s)\, ds, \qquad \varepsilon_t^{y,k} \sim N(0,\, \sigma_y^2). \tag{19}$$

In the shadow-rate framework, the factor loadings are based on the unconstrained forward rate rather than the nominal yield. The unconstrained forward rate follows the Nelson-Siegel structure,

$$f_t(\tau) = B^f(\tau)\, Z_t + A^f(\tau), \qquad B^f(\tau) = \Big[\,1,\ \ e^{-\lambda\tau},\ \ \lambda\tau e^{-\lambda\tau},\ \ 0,\ \ 0\,\Big], \tag{20}$$

where the entries correspond to $L_t^N$, $S_t$, $C_t$, $L_t^R$, and $X_t^{\mathrm{liq}}$, and $A^f(\tau)$ is the forward-rate adjustment term given in (14). The zero entries indicate that the real level factor and the liquidity factor do not enter the nominal pricing equation. Averaging $B^f$ over the interval $[0, \tau_k]$ gives the Nelson-Siegel yield loadings,

$$B^N(\tau_k) = \frac{1}{\tau_k}\int_0^{\tau_k} B^f(s)\, ds = \Big[\,1,\ \ \tfrac{1-e^{-\lambda\tau_k}}{\lambda\tau_k},\ \ \tfrac{1-e^{-\lambda\tau_k}}{\lambda\tau_k} - e^{-\lambda\tau_k},\ \ 0,\ \ 0\,\Big]. \tag{21}$$

These loadings describe the nominal yield only when the unconstrained forward rate is strictly positive across all maturities, in which case $z^N(\cdot)$ reduces to $A^N(\tau_k) + B^N(\tau_k)Z_t$. The presence of $\Phi(\cdot)$ in $\underline{f}_t(s)$ otherwise makes $z^N(\cdot)$ nonlinear in the state vector, with loadings

$$\left.\frac{\partial\, z^N}{\partial\, Z_t}\right|_{\tau_k} = \frac{1}{\tau_k}\int_0^{\tau_k} \Phi\!\left(\frac{f_t(s)}{\omega(s)}\right) B^f(s)\, ds, \tag{22}$$

so that each maturity contributes to the yield loading in proportion to the probability that the unconstrained forward rate is positive at that horizon. The integral is evaluated with the same Riemann sum used for the yield itself.

### C.2.2 TIPS Prices

For TIPS, the model is estimated on individual market prices rather than fitted real yields, which allows the liquidity premium to be identified from the cross-section. Each clean price is scaled by its Macaulay duration $D_t^i$, calculated prior to estimation from market prices and implied yields to maturity, so that pricing errors are comparable across maturities and of the same magnitude as those for nominal yields. Based on equation (16), the measurement equation for the $i$-th TIPS is

$$\frac{P_t^{\mathrm{TIPS},i}(t_0^i,\, t,\, T^i)}{D_t^i} = z^i(Z_t) + \varepsilon_t^{\mathrm{TIPS},i}, \qquad \varepsilon_t^{\mathrm{TIPS},i} \sim N(0,\, \sigma_{\mathrm{TIPS}}^2), \tag{23}$$

and $z^i(\cdot)$ is the nonlinear function

$$z^i(Z_t) = \frac{1}{D_t^i}\sum_{k=1}^{n^i} c_k^i\, \exp\Big\{A^{R,i}(t_0^i, t, \tau_k^i) - \tau_k^i\, B^{R,i}(t_0^i, t, \tau_k^i)\, Z_t\Big\}, \tag{24}$$

and $c_k^i$ denotes the cash flow at payment date $\tau_k^i$. Real yields are not subject to the lower bound and remain affine in the state vector, so the loading vector follows directly from the liquidity-adjusted real yield in equation (11),

$$B^{R,i}(t_0^i, t, \tau_k^i) = \Big[0,\ \ \alpha^R \tfrac{1-e^{-\lambda\tau_k^i}}{\lambda\tau_k^i},\ \ \alpha^R\Big(\tfrac{1-e^{-\lambda\tau_k^i}}{\lambda\tau_k^i} - e^{-\lambda\tau_k^i}\Big),\ \ 1,\ \ \beta^i\,\ell_{\text{liq}}^i(t,\tau_k^i)\Big], \tag{25}$$

$$\ell_{\text{liq}}^i(t,\tau_k^i) = \frac{1-e^{-\kappa_{\text{liq}}^{\mathbb{Q}}\tau_k^i}}{\kappa_{\text{liq}}^{\mathbb{Q}}\tau_k^i} - e^{-\lambda^{L,i}(t-t_0^i)}\frac{1-e^{-(\kappa_{\text{liq}}^{\mathbb{Q}}+\lambda^{L,i})\tau_k^i}}{(\kappa_{\text{liq}}^{\mathbb{Q}}+\lambda^{L,i})\tau_k^i}, \tag{26}$$

where the entries correspond to the same five factors as in equation (20) and $\ell_{\text{liq}}^i(t,\tau_k^i)$ denotes the maturity-specific liquidity loading of bond $i$. The slope and curvature loadings follow the same structure as in $B^N$, scaled by $\alpha^R$, while the nominal level factor is replaced by the real level factor. The bond-specific loading $\beta^i\ell_{\text{liq}}^i$ captures the time-varying liquidity premium of individual TIPS, with the exponential term $e^{-\lambda^{L,i}(t-t_0^i)}$ ensuring that the liquidity component builds up gradually after issuance at $t_0^i$. The exponential discounting of each cash flow at the maturity-specific yield, together with the $\Phi(\cdot)$ term in equation (19), makes the measurement equations nonlinear in $Z_t$, so the standard Kalman filter cannot be applied.

## C.3 Extended Kalman Filter

In the EKF, the nonlinear measurement equations are linearized through a first-order Taylor expansion around the predicted state $Z_{t|t-1}$, so that the standard Kalman filter recursions can be applied. The filter is initialized at the unconditional moments of the state vector,

$$Z_{0|0} = \theta_Z^{\mathbb{P}}, \qquad (\Sigma_{0|0})_{jj} = \frac{\sigma_{jj}^2}{2\kappa_{jj}}. \tag{27}$$

The prediction step projects the filtered state and covariance forward,

$$Z_{t|t-1} = \Phi_0 + \Phi_1\, Z_{t-1|t-1}, \tag{28}$$

$$\Sigma_{t|t-1} = \Phi_1\,\Sigma_{t-1|t-1}\,\Phi_1' + Q. \tag{29}$$

The measurement function $z(\cdot)$, which stacks the nominal yields in equation (19) and the duration-scaled TIPS prices in equation (23), is approximated by a first-order Taylor expansion around $Z_{t|t-1}$,

$$A_t \equiv z(Z_{t|t-1}) - \frac{\partial\, z}{\partial\, Z_t}\bigg|_{Z_{t|t-1}} Z_{t|t-1}, \qquad B_t \equiv \frac{\partial\, z}{\partial\, Z_t}\bigg|_{Z_{t|t-1}},$$

so that the measurement equation takes the locally affine form $y_t = A_t + B_t Z_t + \varepsilon_t$. The Jacobian $B_t$ is evaluated at each filter step and is time varying for both blocks, since the nominal loadings depend on whether the lower bound binds and the TIPS loadings depend on the remaining maturity of each bond. The innovation and its covariance are

$$v_t = y_t - A_t - B_t\, Z_{t|t-1}, \tag{30}$$

$$F_t = B_t\,\Sigma_{t|t-1}\,B_t' + H_t, \tag{31}$$

and the updated state and covariance are

$$Z_{t|t} = Z_{t|t-1} + \Sigma_{t|t-1}\,B_t'\,F_t^{-1}\,v_t, \tag{32}$$

$$\Sigma_{t|t} = \Sigma_{t|t-1} - \Sigma_{t|t-1}\,B_t'\,F_t^{-1}\,B_t\,\Sigma_{t|t-1}. \tag{33}$$

The prediction-error decomposition of the log-likelihood is

$$\log\mathcal{L} = \sum_{t=1}^{T}\left[-\frac{N_t}{2}\log(2\pi) - \frac{1}{2}\log|F_t| - \frac{1}{2}\,v_t'\,F_t^{-1}\,v_t\right], \tag{34}$$

where $N_t$ is the total number of observations at time $t$. The log-likelihood is maximized using the BOBYQA algorithm (Powell, 2009). It is a derivative-free method that converges faster than Nelder-Mead in higher-dimensional settings, and is well suited to our highly nonlinear likelihood surface. BOBYQA is run twice, with the estimates of the first run used as starting values for the second, which resets the trust region and refines the solution.

## C.4 Shifting-Endpoint Specification

The SE-$G^L(7)$ model augments the state vector under $\mathbb{P}$ with the macroeconomic trends from Section 2.2. The $\mathbb{Q}$-dynamics and pricing equations remain unchanged, as only the total level factor enters the pricing equations under $\mathbb{Q}$. The shifting-endpoint extension therefore affects only the $\mathbb{P}$-dynamics, leaving the arbitrage-free pricing relations intact.

### C.4.1 Augmented State Vector

To preserve the orthogonality between trend and cycle innovations required by the unobserved components structure, the state vector is formulated in terms of the cyclical level components $\tilde{L}_t^N = L_t^N - \theta_{1,t}^{\mathbb{P}}$ and $\tilde{L}_t^R = L_t^R - \theta_{4,t}^{\mathbb{P}}$, which are stationary with mean zero,

$$Y_t = \left[\tilde{L}_t^N,\ S_t,\ C_t,\ \tilde{L}_t^R,\ X_t^{\text{liq}},\ r_t^*,\ \pi_t^*\right]' = \left[\tilde{Z}_t',\ \mu_t'\right]',$$

where $\tilde{Z}_t$ collects the five stationary bond-pricing components and $\mu_t = [r_t^*,\ \pi_t^*]'$ the two macroeconomic trends. The effective level factors entering the pricing equations are recovered by adding back the time-varying means in equation (9) to the cyclical level components.

### C.4.2 Discretization

The discrete-time transition equation for $Y_t$ takes the block-diagonal form

$$\begin{pmatrix} \tilde{Z}_t \\ \mu_t \end{pmatrix} = \begin{pmatrix} \Phi_0^{\tilde{Z}} \\ 0 \end{pmatrix} + \begin{pmatrix} \Phi_1^{\tilde{Z}} & 0 \\ 0 & I_2 \end{pmatrix} \begin{pmatrix} \tilde{Z}_{t-1} \\ \mu_{t-1} \end{pmatrix} + \begin{pmatrix} \xi_t \\ \eta_t \end{pmatrix}, \tag{35}$$

where $I_2$ is the identity matrix reflecting the unit-root trend dynamics. The cyclical level components follow stationary AR(1) processes with an unconditional mean of zero. Written out for the augmented state vector, the matrices in equation (35) are

$$\begin{aligned} \Phi_1^{\text{SE}} &= \text{diag}\big(e^{-\kappa_{11}\Delta t},\ e^{-\kappa_{22}\Delta t},\ e^{-\kappa_{33}\Delta t},\ e^{-\kappa_{44}\Delta t},\ e^{-\kappa_{55}\Delta t},\ 1,\ 1\big), \\ \Phi_0^{\text{SE}} &= \left[0,\ (1-e^{-\kappa_{22}\Delta t})\,\theta_2^{\mathbb{P}},\ (1-e^{-\kappa_{33}\Delta t})\,\theta_3^{\mathbb{P}},\ 0,\ (1-e^{-\kappa_{55}\Delta t})\,\theta_5^{\mathbb{P}},\ 0,\ 0\right]', \end{aligned}$$

where the zero entries correspond to the zero-mean cyclical level components and the driftless trend components. The state noise covariance matrix is

$$Q^{\text{SE}} = \text{diag}\big(Q_{11},\ Q_{22},\ Q_{33},\ Q_{44},\ Q_{55},\ \sigma_{r^*}^2\Delta t,\ \sigma_{\pi^*}^2\Delta t\big),$$

where the block-diagonal structure reflects the orthogonality between the stationary innovations $\xi_t$ and the trend innovations $\eta_t$, as required by Watson (1986). This separation ensures that structural changes in underlying macroeconomic fundamentals are attributed to the trend components rather than absorbed by the stationary yield curve factors.

### C.4.3 Trend Measurement Equations

The trends $r_t^*$ and $\pi_t^*$ follow random walks and enter the pricing equations only through the total level factors, so they cannot be separately identified from yields alone. We therefore link them to the observable proxies described in Section 2.2 and treat those proxies as imperfect measures of the underlying trends,

$$\text{HLW}_t = r_t^* + \varepsilon_t^{\text{HLW}}, \qquad \varepsilon_t^{\text{HLW}} \sim N(0, \sigma_{\text{HLW}}^2), \tag{36}$$

$$\text{PTR}_t = \pi_t^* + \varepsilon_t^{\text{PTR}}, \qquad \varepsilon_t^{\text{PTR}} \sim N(0, \sigma_{\text{PTR}}^2), \tag{37}$$

where $\varepsilon_t^{\text{HLW}}$ and $\varepsilon_t^{\text{PTR}}$ are measurement errors with variances $\sigma_{\text{HLW}}^2$ and $\sigma_{\text{PTR}}^2$ estimated jointly with the remaining parameters. The proxies help identify the trend components of the level factors without imposing exact equality, allowing the yield data to influence the trend estimates.

Both proxies are observed quarterly, while yields and TIPS prices are observed monthly. Equations (36) and (37) are therefore appended to the observation vector at quarterly dates only, with $\sigma_{\text{HLW}}^2$ and $\sigma_{\text{PTR}}^2$ added to the diagonal of $H_t$ in those months. In the remaining months the trends are updated through the yield and TIPS data alone. The measurement equations for nominal yields and TIPS prices are unchanged, since the $\mathbb{Q}$-dynamics are identical across the two models.

### C.4.4 Filtering and Likelihood

The stationary factors are initialized at their unconditional moments as in equation (27). The trends do not have a stationary distribution and are initialized at the first available proxy values. To ensure comparability of log-likelihoods across specifications, the proxy log-likelihood contribution $\log \mathcal{L}^{\text{proxy}}$ is subtracted from the joint objective. This decomposition is valid because the proxy measurement errors are independent of the yield and TIPS pricing errors, and the trend innovations $\eta_t$ are orthogonal to the stationary factor innovations $\xi_t$, as reflected by the block-diagonal structure of $Q^{\text{SE}}$. Under these conditions the joint log-likelihood decomposes as

$$\log \mathcal{L}^{\text{SE}} = \log \mathcal{L}^{\text{joint}} - \log \mathcal{L}^{\text{proxy}},$$

so that the reported log-likelihoods measure the fit to bond market data in both models.

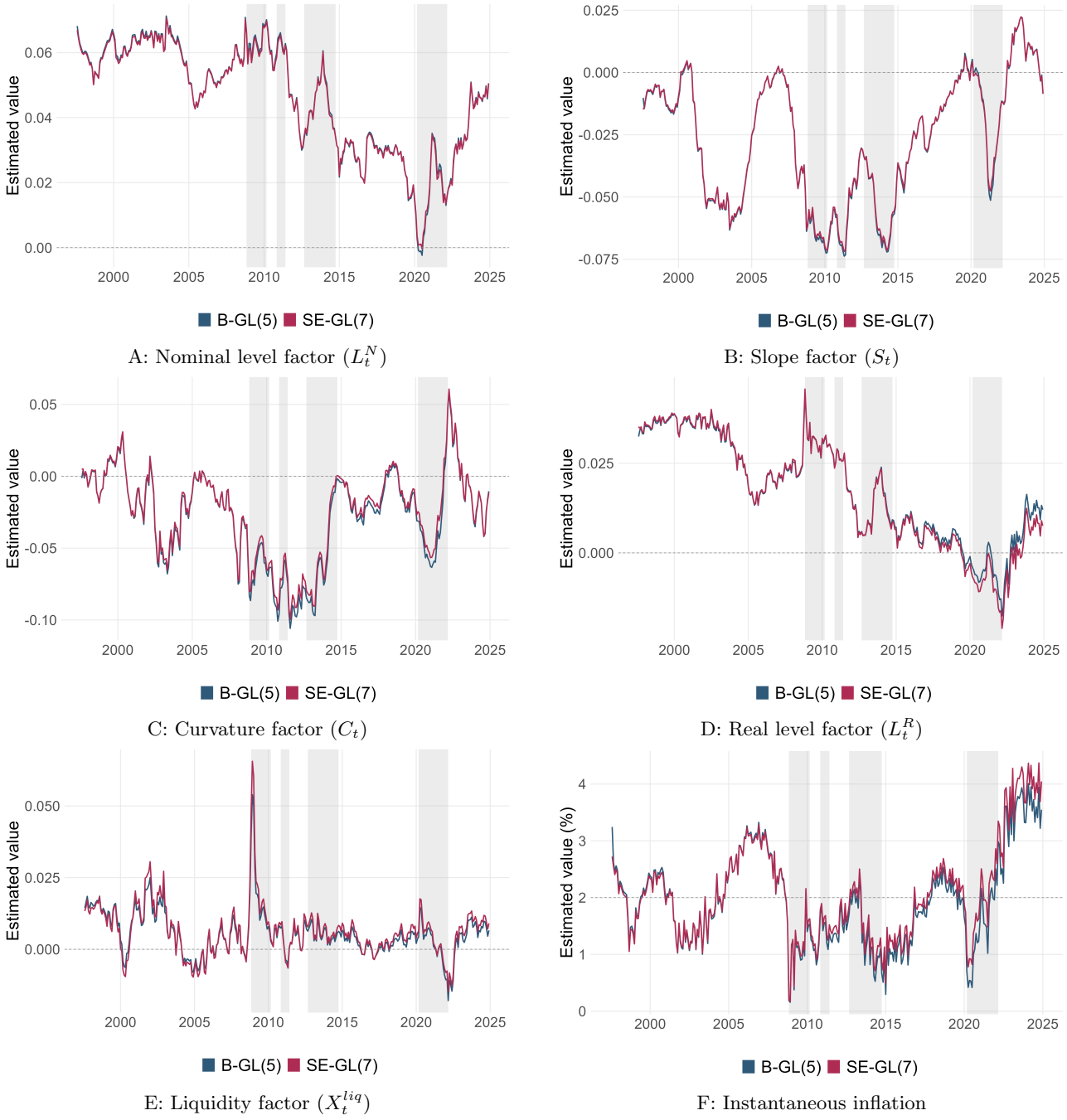
## C.5 Identification and Restrictions

Identification of the liquidity component requires a normalization, as the level of $X_t^{\text{liq}}$ and the bond-specific loadings $\beta^i$ are not jointly identified. Following Andreasen et al. (2021), we set $\beta^1 = 1$ for the first TIPS in the sample and express the remaining loadings relative to this bond.

In the SE-$G^L(7)$ model, upper bounds on the trend innovation volatilities and on the proxy measurement error standard deviations keep the trends from absorbing cyclical variation. The bounds on $\sigma_{\pi^*}$ and $\sigma_{r^*}$ are set at 30 and 52 basis points, roughly 10 basis points above the volatility of the annualized first differences of each proxy series. The bounds on $\sigma_{\text{PTR}}$ and $\sigma_{\text{HLW}}$ are set at 20 and 55 basis points. None of these bounds binds at the optimum.

# Appendix D: Additional Results

## D.1 Factor Paths




A: Nominal level factor ($L_t^N$)

B: Slope factor ($S_t$)

C: Curvature factor ($C_t$)

D: Real level factor ($L_t^R$)

E: Liquidity factor ($X_t^{liq}$)

F: Instantaneous inflation

Figure 4: Estimated paths of the five state variables and instantaneous inflation from August 1997 to December 2024. The instantaneous inflation is computed as the difference between the nominal and real short rate. Shaded areas indicate periods of quantitative easing.

## D.2 Parameter Estimates

Table 5: Estimated Model Parameters: Full Sample

| | B-$G^L(5)$ | | SE-$G^L(7)$ | |
|---|---|---|---|---|
| Parameter | Estimate | SE | Estimate | SE |
| $\kappa_{11}$ | 0.0780 | 0.0179 | 0.2685 | 0.0808 |
| $\kappa_{22}$ | 0.0749 | 0.0069 | 0.1500 | 0.0064 |
| $\kappa_{33}$ | 0.2002 | 0.1094 | 0.2292 | 0.0373 |
| $\kappa_{44}$ | 0.1445 | 0.0919 | 0.3938 | 0.0154 |
| $\kappa_{55}$ | 1.7128 | 0.8152 | 0.8779 | 0.0373 |
| $\sigma_{11}$ | 0.0082 | 0.0004 | 0.0061 | 0.0029 |
| $\sigma_{22}$ | 0.0158 | 0.0015 | 0.0145 | 0.0012 |
| $\sigma_{33}$ | 0.0202 | 0.0009 | 0.0218 | 0.0012 |
| $\sigma_{44}$ | 0.0063 | 0.0003 | 0.0064 | 0.0010 |
| $\sigma_{55}$ | 0.0138 | 0.0014 | 0.0146 | 0.0039 |
| $\theta_1^{\mathbb{P}}$ | 0.0587 | 0.0091 | - | - |
| $\theta_2^{\mathbb{P}}$ | −0.0547 | 0.0086 | −0.0090 | 0.0006 |
| $\theta_3^{\mathbb{P}}$ | −0.0025 | 0.0011 | 0.0040 | 0.0013 |
| $\theta_4^{\mathbb{P}}$ | 0.0182 | 0.0045 | - | - |
| $\theta_5^{\mathbb{P}}$ | 0.0065 | 0.0017 | 0.0083 | 0.0006 |
| $\lambda$ | 0.4777 | 0.0130 | 0.4704 | 0.0149 |
| $\alpha^R$ | 0.6805 | 0.0217 | 0.7016 | 0.0423 |
| $\kappa_{liq}^{\mathbb{Q}}$ | 0.7077 | 0.0465 | 0.8497 | 0.1064 |
| $\theta_{liq}^{\mathbb{Q}}$ | 0.0039 | $1.61\times10^{-4}$ | 0.0039 | $2.75\times10^{-4}$ |
| $\sigma_{\pi^*}$ | - | - | 0.0027 | 0.0002 |
| $\sigma_{r^*}$ | - | - | 0.0047 | 0.0008 |
| $\sigma_{\text{PTR}}$ | - | - | 0.0012 | 0.0010 |
| $\sigma_{\text{HLW}}$ | - | - | 0.0047 | 0.0002 |
| $\sigma_y$ | 0.0006 | $2.84\times10^{-5}$ | 0.0006 | $3.12\times10^{-5}$ |
| $\sigma_{\text{TIPS}}$ | 0.0005 | $3.25\times10^{-5}$ | 0.0005 | $3.12\times10^{-5}$ |

This table reports the parameter estimates and corresponding standard errors for the B-$G^L(5)$ and SE-$G^L(7)$ models, estimated with diagonal $K_Z^{\mathbb{P}}$ and $\Sigma$ matrices. Standard errors are computed using the robust covariance matrix estimator proposed by White (1982).

## D.3 Residual Analysis

Table 6: Normality Tests for Pricing Errors of the SE-$G^L(7)$ Model

| Maturity in months | Skewness | Kurtosis | JB statistic |
|---|---|---|---|
| 3 | −0.50 | 0.44 | 16.37* |
| 6 | −1.67 | 11.45 | 1951.48* |
| 12 | 0.02 | 0.97 | 12.94* |
| 24 | 0.12 | 0.61 | 5.99 |
| 36 | −0.10 | 2.44 | 82.01* |
| 48 | −2.02 | 12.87 | 2493.61* |
| 60 | −1.19 | 5.31 | 463.89* |
| 72 | −1.22 | 4.47 | 355.62* |
| 84 | −0.23 | 2.08 | 62.25* |
| 96 | 0.03 | 3.54 | 171.60* |
| 108 | 0.58 | 14.61 | 2942.80* |
| 120 | 1.38 | 11.65 | 1966.65* |

This table reports normality diagnostics for the nominal yield pricing errors in the SE-$G^L(7)$ model over the period from August 1997 to December 2024. For each maturity, it shows skewness, kurtosis, and the Jarque-Bera test statistic. An asterisk denotes rejection of the null hypothesis of normality at the 5% level.

Table 7: Normality Tests for Individual TIPS Pricing Errors of the SE-$G^L(7)$ Model

| 5-Year TIPS | | | | 10-Year TIPS | | | |
|---|---|---|---|---|---|---|---|
| CUSIP | Skewness | Kurtosis | JB statistic | CUSIP | Skewness | Kurtosis | JB statistic |
| 9128273A8 | 0.23 | 0.03 | 0.32 | 9128272M3 | −0.74 | 1.15 | 13.16* |
| 912828CZ1 | 0.09 | 0.33 | 0.25 | 9128273T7 | −1.75 | 4.40 | 126.33* |
| 912828FB1 | 2.58 | 12.43 | 271.93* | 9128274Y5 | 0.36 | 0.84 | 4.88 |
| 912828GN4 | 3.13 | 10.99 | 239.74* | 9128275W8 | −0.12 | −0.39 | 0.84 |
| 912828HW3 | −0.69 | 3.77 | 24.13* | 9128276R8 | −0.53 | 2.56 | 30.58* |
| 912828KM1 | −0.50 | −0.26 | 1.59 | 9128277J5 | −2.47 | 15.80 | 1095.98* |
| 912828MY3 | 0.10 | −0.62 | 0.64 | 912828AF7 | −3.80 | 25.18 | 2768.59* |
| 912828QD5 | 0.64 | 1.63 | 6.48* | 912828BD1 | −0.38 | 9.64 | 374.02* |
| 912828SQ4 | −0.32 | −0.64 | 1.22 | 912828BW9 | −1.12 | 6.77 | 203.22* |
| 912828UX6 | −1.16 | 1.20 | 10.24* | 912828CP3 | −1.48 | 8.23 | 305.52* |
| 912828C99 | 0.18 | −0.76 | 1.06 | 912828DH0 | −0.03 | 1.80 | 12.93* |
| 912828K33 | −0.75 | 0.41 | 3.66 | 912828EA4 | −0.27 | 2.23 | 21.10* |
| 912828Q60 | 1.49 | 2.69 | 24.20* | 912828ET3 | −0.86 | 2.96 | 46.80* |
| 912828X39 | 2.44 | 9.21 | 162.94* | 912828FL9 | 0.40 | 1.62 | 13.10* |
| 9128284H0 | 1.01 | 2.59 | 16.21* | 912828GD6 | 1.58 | 4.63 | 125.64* |
| 9128286N5 | −2.77 | 10.37 | 207.15* | 912828GX2 | 0.70 | 1.18 | 13.47* |
| 912828YL8 | −0.69 | 0.54 | 3.33 | 912828HN3 | −0.47 | 0.12 | 3.62 |
| 912828ZJ2 | −0.33 | 0.17 | 0.69 | 912828JE1 | 0.49 | 3.40 | 50.20* |
| 91282CAQ4 | 0.22 | −0.56 | 0.75 | 912828JX9 | −0.26 | −0.24 | 1.30 |
| 91282CCA7 | −0.20 | −1.31 | 2.83 | 912828LA6 | 1.58 | 4.60 | 119.39* |
| | | | | 912828MF4 | 0.21 | 0.01 | 0.69 |
| | | | | 912828NM8 | 0.24 | 1.14 | 6.10* |
| | | | | 912828PP9 | 0.32 | 1.42 | 9.76* |
| | | | | 912828QV5 | −0.16 | 0.55 | 1.65 |
| | | | | 912828SA9 | −0.44 | −0.07 | 3.09 |
| | | | | 912828TE0 | −2.14 | 11.93 | 642.61* |
| | | | | 912828UH1 | −0.77 | 0.52 | 10.61* |
| | | | | 912828VM9 | −0.22 | 0.65 | 2.50 |
| | | | | 912828B25 | 0.29 | −0.30 | 1.66 |
| | | | | 912828WU0 | 2.17 | 5.02 | 175.87* |
| | | | | 912828H45 | −0.18 | 0.13 | 0.58 |
| | | | | 912828XL9 | −0.31 | 2.59 | 28.42* |
| | | | | 912828N71 | −0.21 | 1.09 | 5.41 |
| | | | | 912828S50 | 0.06 | 0.41 | 0.74 |
| | | | | 912828V49 | 0.98 | 3.69 | 69.13* |
| | | | | 9128282L3 | 0.32 | 0.36 | 2.05 |
| | | | | 9128283R9 | −0.26 | 0.99 | 4.16 |
| | | | | 912828Y38 | −0.18 | 0.06 | 0.41 |
| | | | | 9128285W6 | −0.24 | 2.00 | 12.56* |
| | | | | 9128287D6 | −0.93 | 1.65 | 16.76* |
| | | | | 912828Z37 | −1.21 | 5.06 | 77.36* |
| | | | | 912828ZZ6 | 1.22 | 4.01 | 48.68* |
| | | | | 91282CBF7 | −0.80 | 2.03 | 13.13* |
| | | | | 91282CCM1 | 0.60 | 3.45 | 22.72* |

This table reports bond-level normality diagnostics for TIPS pricing errors in the SE-$G^L(7)$ model over the period from August 1997 to December 2024. Only TIPS with at least 36 observations are included to ensure a sufficient number of observations for the Jarque-Bera test, while still retaining the 5-year TIPS in the analysis. For each 5-year and 10-year TIPS, it shows the CUSIP, skewness, kurtosis, and the Jarque-Bera test statistic. An asterisk denotes rejection of the null hypothesis of normality at the 5% level.

## D.4 Pricing Errors: Subsample

Table 8: Pricing Errors: Subsample

| | B-$G^L(5)$ | | SE-$G^L(7)$ | |
|---|---|---|---|---|
| Maturity in months | Mean | RMSE | Mean | RMSE |
| Panel A: Nominal yields | | | | |
| 60 | −1.75 | 4.78 | −1.67 | 4.75 |
| 120 | 3.59 | 6.67 | 3.45 | 6.44 |
| All Nominal | −0.38 | 5.89 | −0.38 | 5.78 |
| Panel B: TIPS (averaged by maturity) | | | | |
| 60 | −1.78 | 7.64 | −1.66 | 7.64 |
| 120 | 0.00 | 3.90 | 0.00 | 3.88 |
| All TIPS | −0.25 | 4.61 | −0.24 | 4.60 |
| Log-likelihood | 26104.64 | | 26127.59 | |
| AIC | −52015.28 | | −52057.18 | |
| BIC | −51696.81 | | −51732.14 | |

This table reports mean pricing errors (Mean) and root mean squared errors (RMSE) in basis points for nominal yields and TIPS from August 1997 to December 2013. Results are reported for the B-$G^L(5)$ and SE-$G^L(7)$ models, both estimated with diagonal $K_Z^{\mathbb{P}}$ and $\Sigma$ matrices. The pricing errors are defined as observed minus model-implied yields. For TIPS, market prices are converted to yields to maturity and averaged by maturity group. The final rows report the log-likelihood, Akaike information criterion (AIC), and Bayesian information criterion (BIC).

## D.5 Descriptive Statistics of Level Factor Decomposition

Table 9: Descriptive Statistics for Level Factors

| | Mean | StdDev | Min | Max | $\rho_1$ | $\rho_{12}$ | $\rho_{24}$ | ADF |
|---|---|---|---|---|---|---|---|---|
| Panel A: Nominal Level Factor | | | | | | | | |
| Total | 0.0456 | 0.0167 | −0.0004 | 0.0706 | 0.983 | 0.781 | 0.681 | −1.971 |
| Trend | 0.0445 | 0.0117 | 0.0235 | 0.0663 | 0.989 | 0.878 | 0.782 | −1.689 |
| Cycle | 0.0011 | 0.0100 | −0.0266 | 0.0244 | 0.977 | 0.693 | 0.434 | −1.612 |
| Panel B: Real Level Factor | | | | | | | | |
| Total | 0.0170 | 0.0155 | −0.0211 | 0.0457 | 0.984 | 0.849 | 0.723 | −1.584 |
| Trend | 0.0208 | 0.0113 | 0.0025 | 0.0425 | 0.992 | 0.896 | 0.797 | −1.359 |
| Cycle | −0.0038 | 0.0085 | −0.0301 | 0.0143 | 0.971 | 0.754 | 0.453 | −1.486 |

This table reports descriptive statistics and autocorrelations at lags 1, 12 and 24 for the nominal and real level factors and their trend and cycle components. Stationarity is assessed using the Augmented Dickey-Fuller (ADF) test, with the lag length selected based on the Akaike Information Criterion (AIC) and equal to one for all series. The critical values for the ADF test are taken from MacKinnon (1996). In the ADF tests, an intercept is included for the total level factor and the trend component, whereas it is excluded for the cycle components.

# Appendix E: Forecasting

## E.1 Five-Year Yield

Table 10: Out-of-sample Performance for Five-year Yields

| | Forecast Horizon (in months) | | | |
|---|---|---|---|---|
| | 6 (T = 126) | 12 (T = 120) | 18 (T = 114) | 24 (T = 108) |
| Panel A: Nominal five-year yield | | | | |
| B-$G^L(5)$ | 65.41 | 104.93 | 138.99 | 168.58 |
| SE-$G^L(7)$ | **63.37**$^*$ | **98.76**$^*$ | **127.16**$^*$ | **151.45**$^*$ |
| RW | 65.73 | 107.13 | 144.84 | 178.91 |
| Panel B: Real five-year yield | | | | |
| B-$G^L(5)$ | 61.78 | 101.74 | 139.11 | 166.49 |
| SE-$G^L(7)$ | **58.85** | **87.63**$^*$ | **112.90**$^*$ | **131.78**$^*$ |
| RW | 64.77 | 107.89 | 147.28 | 176.36 |

This table reports root mean squared forecast errors in basis points for forecasts of nominal and real five-year yields from January 2014 to December 2024. The models are B-$G^L(5)$, SE-$G^L(7)$, and a random walk benchmark (RW), evaluated at horizons $h = 6, 12, 18,$ and 24 months. The lowest RMSFE within each panel is highlighted in bold. An asterisk (*) indicates that SE-$G^L(7)$ significantly outperforms B-$G^L(5)$ at the 5% level, based on a one-sided Diebold-Mariano test with the Harvey et al. (1997) small-sample correction and Newey-West standard errors with $h - 1$ lags.

## E.2 COVID Exclusion

Table 11: Out-of-sample Performance for Ten-year Yields Excluding the COVID Period

| | Forecast Horizon (in months) | | | |
|---|---|---|---|---|
| | 6 (T = 107) | 12 (T = 95) | 18 (T = 88) | 24 (T = 82) |
| Panel A: Nominal ten-year yield | | | | |
| B-$G^L(5)$ | 57.72 | 85.63 | 98.30 | 112.72 |
| SE-$G^L(7)$ | **54.59**$^*$ | **77.97**$^*$ | **85.96**$^*$ | **96.73**$^*$ |
| RW | 59.03 | 90.63 | 105.04 | 119.43 |
| Panel B: Real ten-year yield | | | | |
| B-$G^L(5)$ | 62.09 | 83.95 | 105.80 | 128.72 |
| SE-$G^L(7)$ | 56.95$^*$ | **71.09**$^*$ | **81.86**$^*$ | **93.14**$^*$ |
| RW | **51.61** | 83.82 | 107.06 | 123.67 |

This table reports root mean squared forecast errors in basis points for forecasts of nominal and real ten-year yields from January 2014 to December 2024, excluding the COVID period. Specifically, the period from March 2020 to March 2021 is omitted. The models are B-$G^L(5)$, SE-$G^L(7)$, and a random walk benchmark (RW), evaluated at horizons $h = 6, 12, 18,$ and 24 months. The lowest RMSFE within each panel is highlighted in bold. An asterisk (*) indicates that SE-$G^L(7)$ significantly outperforms B-$G^L(5)$ at the 5% level, based on a one-sided Diebold-Mariano test with the Harvey et al. (1997) small-sample correction and Newey-West standard errors with $h - 1$ lags.